

Assignment 3 Report: Web Design with Bootstrap 5 and Media Queries

Course / Module: Front-End Web Development

Project Title: Multi-Device Adaptive Interface Implementation

Date: September 29, 2026

Live Site URL: <https://dazzling-paprenjak-589e9f.netlify.app/>

1. Executive Summary

This project demonstrates the implementation of a fully responsive, accessible, and modern web application. By combining custom CSS media queries with the Bootstrap 5 front-end framework, the application delivers an optimal user experience across mobile, tablet, and desktop devices. The implementation strictly adheres to modern HTML5 semantic standards and WCAG accessibility guidelines.

2. Project Objectives

1. Implement responsive typography and custom layouts using pure CSS Media Queries.
2. Utilize Bootstrap 5's 12-column grid system (containers, rows, and responsive cols) for structured page layouts.
3. Integrate core Bootstrap UI components, including Navigation Bars, Carousels, Card groups, and Forms.
4. Apply responsive utility classes (spacing, flexbox, display) to manage spacing and alignment dynamically.
5. Ensure accessibility (a11y) using semantic HTML5 tags and appropriate ARIA attributes.

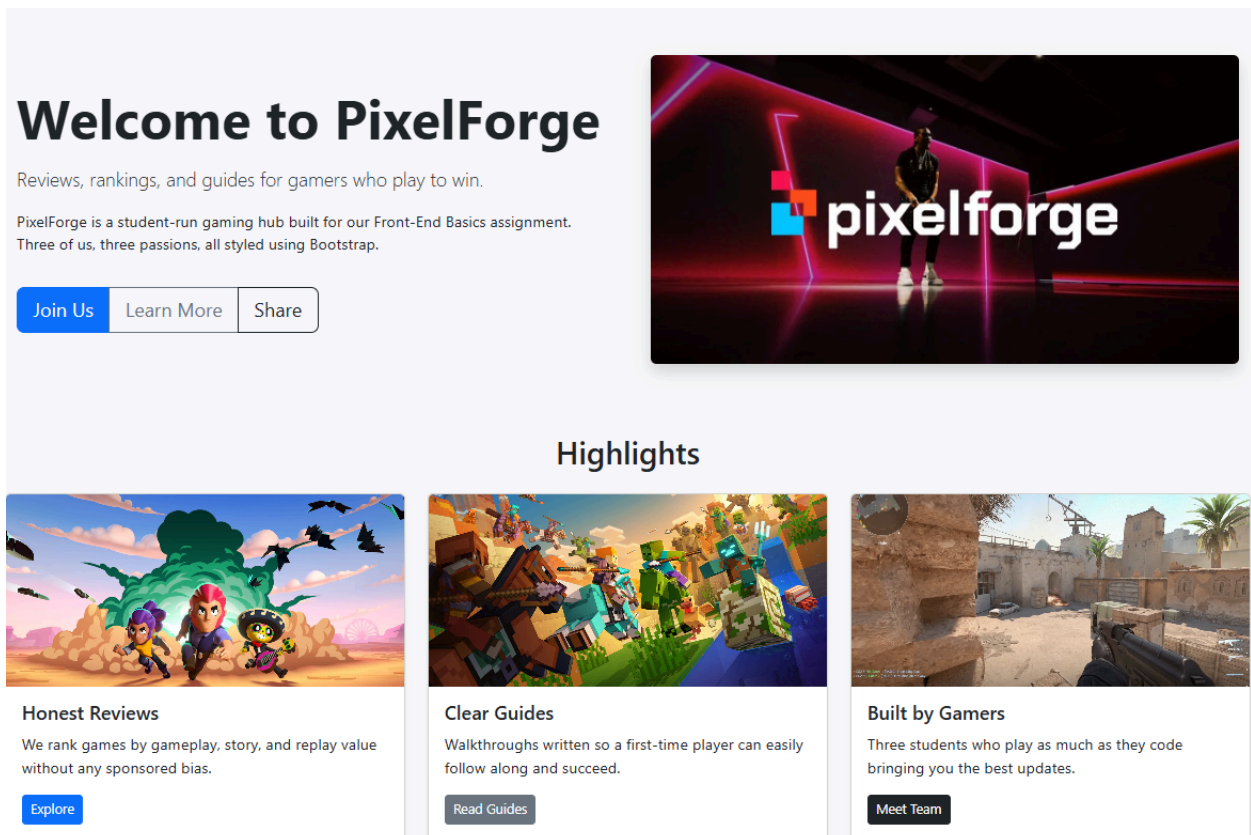
3. Implementation Details & Architecture

Part 1: Custom CSS Media Queries

Responsive Typography (Task 1):

1. Implemented dynamic scaling for headings (`.responsive-heading`) and paragraphs (`.responsive-paragraph`).
2. Defined custom breakpoints:

- a. **Mobile (< 768px):** Font size scaled for compact viewports.
- b. **Tablet (768px – 1023px):** Medium scaling.
- c. **Desktop (≥ 1024px):** Large viewport optimization.
- **Pure CSS Layout Grid (Task 2):**
 - 1. Built a flexbox-based card layout (`.custom-card-group`) independent of framework grid systems.
 - 2. Configured responsive column wrapping:
 - a. **Mobile:** Single-column layout (100% width).
 - b. **Tablet:** Two-column grid (50% width per item).
 - c. **Desktop:** Three-column layout using flexible growing factors (`flex: 1`).



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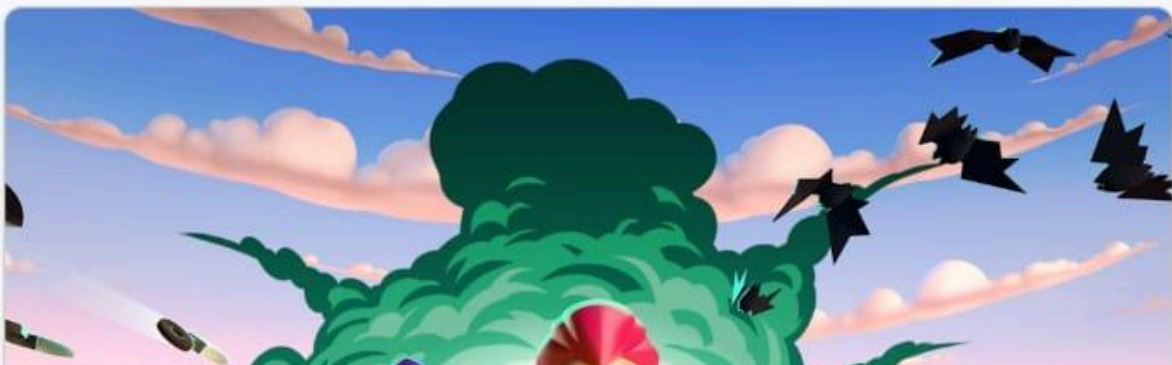
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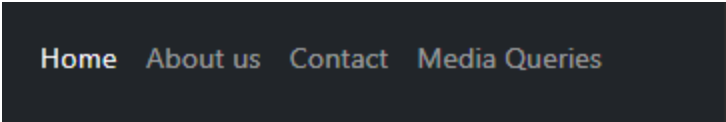
Part 2: Bootstrap 5 Integration & UI Components

Grid System & Layout (Task 3):

1. Implemented a two-column layout (`col-12 col-lg-6`) for primary content blocks on desktop, collapsing to full width on smaller screens.
2. Structured a three-column layout (`col-12 col-md-6 col-lg-4`) for featured content cards.

Responsive Navigation Bar (Task 5 & Task 10):

1. Created a sticky navigation bar (`.sticky-top`) wrapped in a semantic `<nav>` element.
2. Integrated a collapsible menu (`navbar-toggler`) driven by Bootstrap's JavaScript bundle for mobile viewports.



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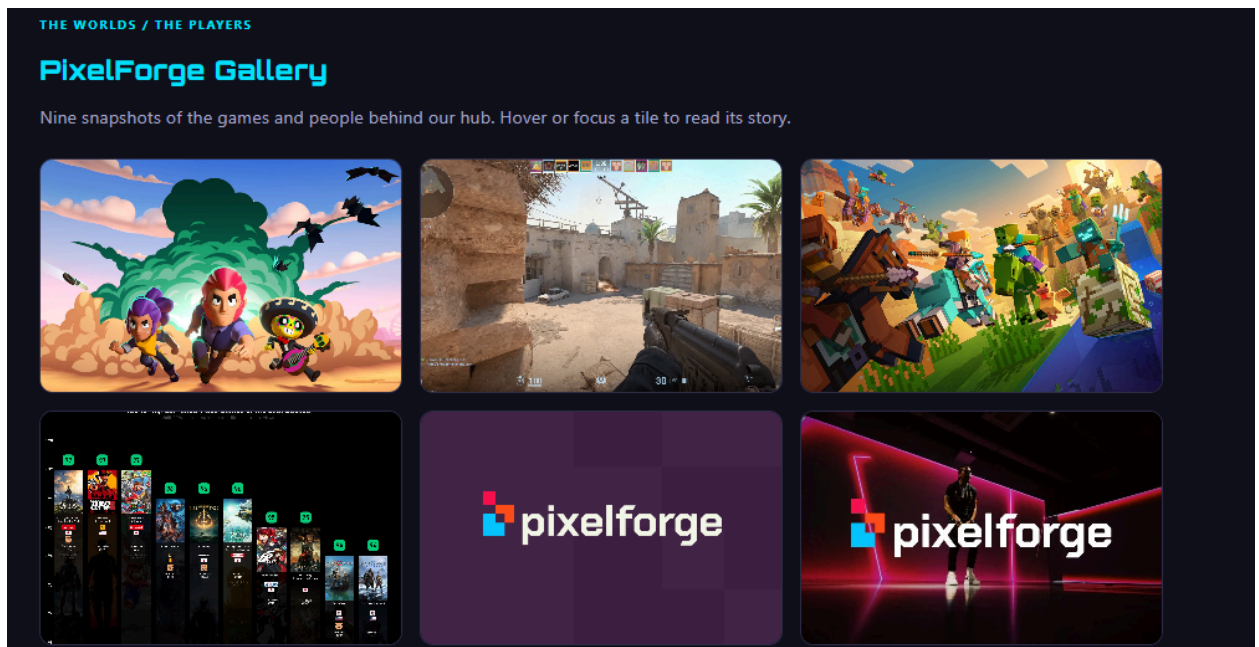
Media Queries



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Interactive Components (Tasks 6, 7 & 8):

- Buttons:** Implemented standard button variants (`btn-primary`, `btn-outline-success`), sizing variations (`btn-lg`, `btn-sm`), and an organized button group (`btn-group`).
- Carousel:** Built a multi-item slideshow featuring 6 responsive images with explicit slide indicators and control controls (`carousel-control-prev/next`).
- Cards:** Applied card components with matching heights (`h-100`) and shadow utilities (`shadow-sm`) to ensure consistent visual alignment across rows.



Forms & Utility Classes (Tasks 4 & 9):

- Form inputs wrapped in clean form-group structures with custom styling and input groups (`input-group-text`).

2. Applied spacing utilities across breakpoints (e.g., `mb-5`, `p-4`, `mt-lg-3`, `px-sm-2`) to avoid hardcoded CSS margins where appropriate.

The image shows a 'Contact Us' form with the following fields and elements:

- Full Name:** A text input field with the placeholder text 'Enter your full name'.
- Email address:** A text input field containing 'name@example.com'. It has a red error icon and a tooltip that says 'Заполните это поле.' (Fill in this field).
- Reason for Contact:** A dropdown menu with the text 'Open this select menu' and a downward arrow.
- Preferred Contact Method:** Two radio buttons labeled 'Email' (selected) and 'Phone'.
- Subscribe to newsletter:** A checkbox.
- Message:** A large text area for the user's message.
- Submit Form:** A blue button at the bottom of the form.

Part 3: Accessibility & Semantic Compliance (Task 10)

Semantic HTML5: Built using structure tags (`<nav>`, `<main>`, `<section>`, `<header>`, `<footer>`).

ARIA & Accessibility Standards:

1. Added `aria-label` and `aria-controls` to interactive navigation toggles.
2. Applied `aria-current="page"` to active links.
3. Included descriptive `alt` tags on all 9 carousel images and card media to support screen-reader navigation.
4. Form inputs are explicitly paired with corresponding `<label>` elements using matching `for/id` attributes.

4. Verification & Testing Matrix

Feature / Screen Size	Mobile (< 768px)	Tablet (768px - 1023px)	Desktop (≥ 1024px)	Status
Typography	Compact scaling	Medium scaling	Large viewport optimization	Passed
Custom CSS Cards	1 Column	2 Columns	3 Columns	Passed
Bootstrap Grid	Collapsed (1 Col)	Mixed (2 Cols)	3 Columns	Passed
Navbar	Hamburger Menu	Hamburger / Expanded	Fully Expanded	Passed

Carousel Controls	Touch / Tap	Arrow / Touch	Arrow / Click	Passed
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5. Reflection & Conclusion

The combination of standard Bootstrap 5 components and custom media queries proves to be an efficient methodology for modern front-end web development. Bootstrap accelerates the creation of complex UI structures—such as carousels, responsive grids, and forms—while custom CSS media queries provide precise control over typography and specialized design components.